

Theory of Rigid Body Motion and Linearized Error Dynamics

1 Introduction

This document outlines the theoretical foundations for the MATLAB script `example_torque_free_motion.m`. The script simulates the rotational dynamics of a rigid body (a CubeSat) subject to zero external torques. It propagates a nominal trajectory and compares it against slightly perturbed trajectories using both the full nonlinear equations of motion and a linearized error model. The same framework also extends naturally to the case where an external torque depends on attitude angle.

2 Nonlinear Equations of Motion

The rotational state of the rigid body is defined by its attitude quaternion q (representing the rotation from the inertial frame to the body frame) and its angular velocity ω expressed in the body frame.

2.1 Attitude Kinematics

We use the scalar-first quaternion convention $q = [q_0; \vec{q}_v]^T$. The kinematic differential equation governing the evolution of the attitude quaternion is:

$$\dot{q} = \frac{1}{2}q \otimes \begin{bmatrix} 0 \\ \omega \end{bmatrix} \quad (1)$$

where $\otimes$ denotes quaternion multiplication.

2.2 Rigid Body Dynamics (Euler's Equations)

The rotational dynamics of a rigid body are governed by Euler's equation. In the most general case considered here,

$$J\dot{\omega} + \omega \times (J\omega) = \tau_{ext}(q, \omega, t) \quad (2)$$

where J is the principal moment of inertia matrix of the body and τ_{ext} is the applied external torque, expressed in the body frame.

For the torque-free script, $\tau_{ext} = 0$, so the equation reduces to:

$$J\dot{\omega} + \omega \times (J\omega) = 0 \quad \implies \quad \dot{\omega} = -J^{-1}(\omega \times (J\omega)) \quad (3)$$

If a torque depends on attitude angle, then the nonlinear dynamics simply keep the same structure with a nonzero right-hand side. Typical examples are gravity-gradient torques, magnetic torques, spring-like restoring torques, or control torques generated from an attitude error law.

3 Attitude Error Representation

To study perturbations, we define a nominal trajectory $(q_{nom}(t), \omega_{nom}(t))$ and a perturbed trajectory $(q_{pert}(t), \omega_{pert}(t))$.

The attitude error between the nominal body frame and the perturbed body frame is represented by the error quaternion q_{err} , defined via the multiplicative attitude error:

$$q_{err} = q_{nom}^{-1} \otimes q_{pert} \quad (4)$$

For small angular deviations, the error quaternion can be approximated as:

$$q_{err} \approx \begin{bmatrix} 1 \\ \frac{1}{2}\delta\theta \end{bmatrix} \quad (5)$$

where $\delta\theta \in \mathbb{R}^3$ is the small-angle error vector (Euler angles) relating the nominal and perturbed body frames. Therefore, the attitude error vector can be extracted as $\delta\theta \approx 2\vec{q}_{err,v}$.

The angular rate error is simply defined as the difference between the perturbed and nominal angular rates:

$$\delta\omega = \omega_{pert} - \omega_{nom} \quad (6)$$

4 Linearized Error Dynamics

To analyze stability and design linear controllers, we linearize the nonlinear kinematics and dynamics about the time-varying nominal trajectory $(\omega_{nom}(t))$.

4.1 Linearized Kinematics

Taking the derivative of the attitude error vector $\delta\theta$ expressed in the rotating nominal body frame yields an extra Coriolis-like term:

$$\delta\dot{\theta} = \delta\omega - \omega_{nom} \times \delta\theta \quad (7)$$

This term accounts for the relative rotation of the reference frame in which the attitude error is measured.

4.2 Linearized Dynamics

For the torque-free case, we substitute $\omega = \omega_{nom} + \delta\omega$ into Euler's equation:

$$J(\dot{\omega}_{nom} + \delta\dot{\omega}) = -(\omega_{nom} + \delta\omega) \times J(\omega_{nom} + \delta\omega) \quad (8)$$

Expanding the cross products and noting that $J\dot{\omega}_{nom} = -\omega_{nom} \times J\omega_{nom}$:

$$J\delta\dot{\omega} \approx -\delta\omega \times (J\omega_{nom}) - \omega_{nom} \times (J\delta\omega) \quad (9)$$

where we have dropped the second-order term $\delta\omega \times (J\delta\omega)$. Using the anti-commutative property of the cross product ($a \times b = -b \times a$), we can rewrite $-\delta\omega \times (J\omega_{nom})$ as $(J\omega_{nom}) \times \delta\omega$. This yields the linearized dynamics implemented in the script:

$$\delta\dot{\omega} = J^{-1} \left[(J\omega_{nom}) \times \delta\omega - \omega_{nom} \times (J\delta\omega) \right] \quad (10)$$

Using the skew-symmetric matrix operator $[\cdot]^\times$, this can be written as:

$$\delta\dot{\omega} = J^{-1} \left[[J\omega_{nom}]^\times \delta\omega - [\omega_{nom}]^\times J\delta\omega \right] \quad (11)$$

4.3 If the Torque Depends on Angle

Now suppose the external torque depends on attitude, or more generally on attitude, rate, and time:

$$J\dot{\omega} + \omega \times (J\omega) = \tau_{ext}(q, \omega, t) \quad (12)$$

Let the nominal trajectory satisfy

$$J\dot{\omega}_{nom} + \omega_{nom} \times (J\omega_{nom}) = \tau_{nom} \quad (13)$$

with $\tau_{nom} = \tau_{ext}(q_{nom}, \omega_{nom}, t)$. Linearizing the torque term gives

$$\delta\tau \approx K_\theta(t) \delta\theta + K_\omega(t) \delta\omega \quad (14)$$

where the Jacobian matrices are local sensitivity matrices of torque with respect to attitude error and rate error along the nominal trajectory.

The linearized rotational dynamics become

$$J\delta\dot{\omega} = [J\omega_{nom}]^\times \delta\omega - [\omega_{nom}]^\times J\delta\omega + \delta\tau \quad (15)$$

or, after substituting the local torque model,

$$\delta\dot{\omega} = J^{-1} \left([J\omega_{nom}]^\times \delta\omega - [\omega_{nom}]^\times J\delta\omega + K_\theta \delta\theta + K_\omega \delta\omega \right) \quad (16)$$

while the kinematics remain

$$\delta\dot{\theta} = \delta\omega - \omega_{nom} \times \delta\theta \quad (17)$$

Therefore, an angle-dependent torque appears in the linearized model as an additional stiffness-like term $K_\theta \delta\theta$.

4.4 Important Special Case: Restoring Torque

If the nominal motion is an equilibrium with $\omega_{nom} = 0$ and the torque is a restoring torque near that equilibrium, then a common local model is

$$\tau_{ext}(\theta) \approx -K \delta\theta \quad (18)$$

where K is an attitude stiffness matrix. The linearized equations reduce to

$$\delta\dot{\theta} = \delta\omega \quad (19)$$

$$J\delta\dot{\omega} = -K \delta\theta \quad (20)$$

Combining the two gives the familiar second-order small-angle system

$$J\delta\ddot{\theta} + K \delta\theta = 0 \quad (21)$$

which is the rotational analog of a mass-spring oscillator.

If damping or active rate feedback is present, for example $\tau_{ext} \approx -K \delta\theta - C \delta\omega$, then

$$J\delta\ddot{\theta} + C \delta\dot{\theta} + K \delta\theta = 0 \quad (22)$$

which is the rotational analog of a damped mass-spring system.

4.5 Nonlinear Angle-Dependent Torques

If the exact torque is nonlinear in angle, for example

$$\tau_{ext}(\theta) = -k \sin \theta \quad (23)$$

then the full nonlinear propagation must retain that exact torque law. The small-angle linearization replaces $\sin \theta$ with θ , giving

$$\tau_{ext}(\theta) \approx -k\theta \quad (24)$$

which is accurate only while the attitude error remains small.

5 Conservation Laws

In the absence of external torques, two key quantities remain strictly conserved throughout the motion:

1. **Angular Momentum:** The angular momentum vector $\vec{H} = J\omega$ is constant in the inertial frame. Consequently, its magnitude in the body frame, $\|\vec{H}\| = \|J\omega\|$, remains constant.
2. **Rotational Kinetic Energy:** The rotational kinetic energy $T_{rot} = \frac{1}{2}\omega^T J\omega$ is constant.

The script verifies the numerical accuracy of the propagation by checking the drift of these two conserved quantities over time.

With a nonzero external torque, these conservation laws change:

- Angular momentum is generally no longer constant, because the external torque changes it according to $\dot{\vec{H}} = \tau_{ext}$ in the inertial frame.
- Rotational kinetic energy is generally no longer constant either, because the torque can do work on the body.
- If the torque is conservative and derives from a potential $U(\theta)$, then total mechanical energy,

$$E = \frac{1}{2}\omega^T J\omega + U(\theta) \quad (25)$$

is conserved in the absence of damping and explicit time dependence.

So, for the current script, $\tau_{ext} = 0$ and the motion is torque-free. If you add a torque that depends on angle, the nonlinear dynamics gain a forcing term and the linearized error model gains the additional stiffness-like contribution described above.